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Chemical remanent magnetization due to deep-burial diagenesis in oolitic hematite-bearing ironstones of Alabama

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Oolitic hematite-bearing ironstones of the Silurian Red Mountain Formation of Alabama are shown to carry a single-component remanence stable enough to have survived major folding (of probable Permian age). Nevertheless, the remanence direction (ten sites yielding a paleopole at 38.0°N, 132.4°E with $dm = 3.6°$, $dp = 1.9°$), its reverse polarity and a negative intraformational conglomerate test show that the remanence was very likely acquired during the Pennsylvanian—some 130 Ma after deposition. This remanence is likely a chemical remanent magnetization (CRM) acquired during diagenesis induced by heating due to deep burial under a Pennsylvanian clastic wedge. Two possible mechanisms for acquisition of CRM during deep-burial diagenesis are considered. In hypothesis I, the oolitic hematite transformed from original goethite when heated to about 80°C, acquiring CRM. In hypothesis II, the oolitic hematite originated from ferrihydrite and was too fine-grained to acquire stable CRM until heat raised the solubility of hematite allowing grain growth. Hypothesis I explains the timing of remanence acquisition better, but there is some evidence that oolitic goethites may be stable to considerably more than 80°C. Hypothesis II has some difficulty explaining preliminary paleomagnetic results from oolitic hematite-bearing ironstones of the Silurian Clinton Group, New York State. We prefer hypothesis I but both hypotheses remain plausible. Both hypotheses warn that continental red beds may also acquire CRM during diagenesis induced by deep-burial heating, long after deposition but before folding.

1. Introduction

Continental red beds have often been studied paleomagnetically. However, the hematite in red beds frequently carries a multicomponent remanence probably acquired during long-continuing diagenesis. Diagenesis occurs at low temperature ($< 200°C$ according to Fairbridge, 1983) but can produce hematite with stable chemical remanence —often from dissolution of iron-bearing silicates long after deposition (Turner, 1980).

Marine red beds are far less common but include oolitic hematite-bearing ironstones, mostly of Early Paleozoic age, which have been an important source of iron ore. Oolitic hematite-bearing ironstones could be less subject to the remagnetizations common in continental red beds since the high hematite content of the ironstones should make secondary hematite from dissolution of iron silicates relatively unimportant.

When we began this study, only five oolitic hematite-bearing ironstones had been studied paleomagnetically—the iron ores of Birmingham Alabama (Howell et al., 1958, 1960) three iron ores from Czechoslovakia (Andreeva et al., 1965) and the iron ores of Bell Island, Newfoundland (Buchan and Hodych, 1982a). Only the ores from Newfoundland had been reported in enough detail

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to be sure of a stable single-component remanence. Unfortunately, no fold test was possible so even this remanence could have been acquired long after Early Ordovician deposition during Devonian orogeny.

We began the present study of the Silurian oolitic hematite ores of Birmingham Alabama because of their similarity to the ores of Bell Island, Newfoundland and because a fold test was possible. Besides, only two reliable Silurian paleopole determinations had been published in full for North America excluding paleopoles from suspect displaced terraines (Irving, 1979). The original magnetic study of Birmingham ores (Howell et al., 1958, 1960) concentrated on magnetic anisotropy and was not used to define a Silurian paleopole because the samples were only from one site and no thermal demagnetization was carried out. Furthermore, the remanence was of shallow inclination and directed to the south which seems characteristic of Pennsylvanian or Permian post-folding remagnetization farther north in the Valley and Ridge Province (Van der Voo, 1979). Hence a test of whether the magnetization was acquired before or after folding was crucial.

We presented preliminary paleomagnetic results for the oolitic hematites of Birmingham at the spring 1982 meeting of the American Geophysical Union (Buchan and Hodych, 1982b) reporting a stable single-component remanence acquired before folding of the Birmingham anticlinorium. We presented further results at the fall 1983 meeting of the American Geophysical Union (Hodych et al., 1983) and in a short communication (Hodych et al., 1984) finding from a negative conglomerate test and from the direction and polarity of remanence that the Birmingham oolitic hematites were very likely magnetized during the Pennsylvanian —some 130 Ma after their Silurian deposition. Similar conclusions were presented at the spring 1983 meeting of the American Geophysical Union by Van der Voo et al. (1983) and recently appeared in a paper by Perroud and Van der Voo (1984) brought to our attention while our present paper was being reviewed.

In this paper, we give details of our paleomagnetic results on the Birmingham oolitic hematites and concentrate on the process by which they acquired a stable single-component remanence in the Pennsylvanian long after deposition but shortly before folding. We have already pointed out (Hodych et al., 1983, 1984) that this behavior is understandable if the ores originated as oolitic goethite and required heating to ~ 80°C to transform to hematite and acquire stable chemical remanent magnetization (CRM). Such temperatures probably did not occur until deep Pennsylvanian burial. This hypothesis is tested and remains plausible though it encounters some difficulty from evidence that significantly more than 80°C may be required to transform oolitic goethite to hematite. An alternative hypothesis is considered in which the hematite was originally very fine-grained, originating from a short-lived ferrihydrite precursor; the hematite coarsened enough to acquire stable CRM during heating due to deep Pennsylvanian buria.l. This second hypothesis is more difficult to test but does encounter difficulty in explaining our preliminary paleomagnetic results from oolitic hematite-bearing ironstones of New York State. Both hypotheses are also relevant to the paleomagnetism of continental red beds since their hematite pigment may have had a goethite or a ferrihydrite precursor. Both hypotheses show the need for caution in interpreting the paleomagnetism of red beds since diagenetic changes during burial heating may produce CRM long after deposition but before folding.

Planning and interpretation in this paper were by J.P.H. Collecting and most measurements were by R.R.P. Data analysis and initial measurements were by K.L.B.

2. Samples and sampling sites

Marine red beds of the Red Mountain Formation of Silurian (Llandoverian) age were sampled near Birmingham Alabama at the southern end of the folded Valley and Ridge Province of the Appalachians (see inset to Fig. 1). Both limbs of the Birmingham anticlinorium were sampled (Fig. 1 and Table I) to provide a fold test of timing of remanence acquisition. Also, eleven cobbles in an intraformational conglomerate within the Red Mountain Formation were sampled at site 1d to

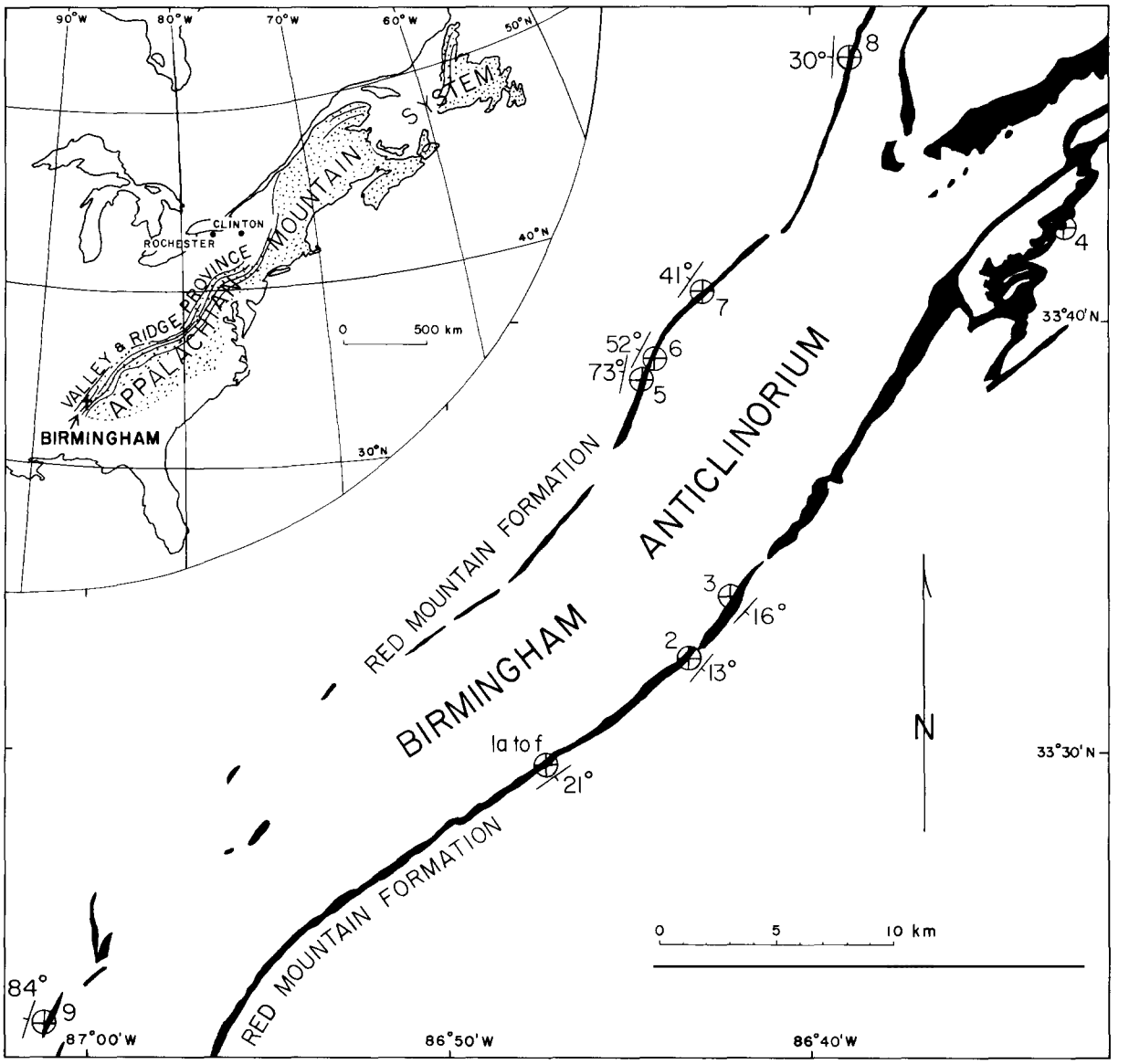


Fig. 1. Paleomagnetic sampling sites (renumbered from Hodych et al., 1984) in the Silurian Red Mountain Formation (outcrop in solid black after Burchard and Butts, 1910) near Birmingham Alabama. Sites lie on both limbs of the Birmingham anticlinorium in the Valley and Ridge Province of the southern Appalachian Mountains. The inset shows the Appalachian Mountain system dotted with the trend of fold axes in foreland sediments—including those of the Valley and Ridge Province—indicated by lines (after King, 1977).

provide a conglomerate test of timing of remanence acquisition.

The stratigraphic level within the Red Mountain Formation was only known at sites 1a, 1b, 1c, 1d, 1e and 1f—all in the Red Mountain Expressway roadcut. Here the Red Mountain Formation is ~ 70 m thick consisting of interbedded sandstone and siltstone with four prominent intervals of red sedimentary ironstone (Chowns and McKinney, 1980).

TABLE I

Sampling sites in the Red Mountain Formation

Site Number	Location		Tilt of bedding		Rock description	Average density (g cm^{-3})	J_N **
	Latitude N	Longitude W	Strike	Dip			
SE limb of Birmingham anticlinorium							
1a	33°29.7′	86°47.3′	57°	20°E	* Upper Red Mtn. Fm. red ironstone	2.9	7.7
1b	33°29.7′	86°47.3′	57°	20°E	Upper Red Mtn. Fm. light grey siltstone	2.6	0.4
1c	33°29.7′	86°47.3′	54°	21°E	* Middle Red Mtn. Fm. red ironstone	3.2	9.1
1d	33°29.7′	86°47.3′	54°	21°E	* Middle Red Mtn. Fm. (base) calcareous red ironstone cobbles in conglomerate	2.9	3.9
1e	33°29.7′	86°47.3′	54°	21°E	* Lower Red Mtn. Fm. (top) red ironstone	3.3	12.6
1f	33°29.7′	86°47.3′	45°	20°E	* Lower Red Mtn. Fm. reddish grey sandstone	2.7	3.6
2	33°32.2′	86°43.4′	34°	13°E	* red ironstone	2.9	13.9
3	33°33.6′	86°42.2′	40°	16°E	* red sandstone	2.7	7.9
4	33°42.2′	86°33.0′	0°	0°	yellowish sandstone	2.2	1.2
NW limb							
5	33°38.7′	86°44.6′	185°	73°W	* red ironstone	2.9	8.7
6	33°39.4′	86°44.3′	207°	52°W	* reddish grey sandstone	2.4	5.9
7	33°40.7′	86°43.0′	214°	41°W	*red ironstone	2.8	8.1
8	33°6.2′	86°38.9′	360°	30°W	red sandstone	2.2	1.5
9	33°23.6′	87°1.2′	195°	*** 84°W	* red ironstone	3.4	13.5

* Contains oolitic hematite.

** J_N is the average NRM intensity in units of 10^{-5} emu cm^{-3}.

*** At site 9, tilt correction in doubt because of possible rotation in plane of bedding.

With the exception of samples at sites 1b and 4 (Table I), all samples collected were red due to the presence of hematite (though samples at sites 1f and 6 were dark grey with only a slight reddish colour). Most samples collected were sandy sediments—the exceptions being at site 1b and at site 1d. The samples at most sites could probably be classed as ironstones (defined as sediments containing at least 15% by weight of iron). This is supported by density determinations; 15% by weight of iron in the form of hematite should increase the density of sandstone from ~ 2.4 g cm^{-3} to ~ 2.7 g cm^{-3} and indeed at most sites average density was 2.7 g cm^{-3} or greater (Table I). The samples at sites 1b, 4, 6, 8 and probably 1f and 3 should not be classed as ironstones judging by density and appearance.

Twenty polished sections were examined—at least one per site. Sections from all sites except 1b, 4 and 8 show the presence of oolitic hematite. True ooliths occur consisting of concentric layers of hematite often with a sand grain as center. But hematite occurring as coatings on fossil fragments as at site 1c is also of oolitic structure as pointed out by Alling (1947) for the Silurian hematite ores of New York State. Hematite replacement of fossil fragments and hematite cement are also seen in the ironstones. The finest-grained sandstones at sites 1f and 6 do not seem to contain hematite ooliths but may contain detritus from hematite ooliths.

At all sites where oolitic hematite is present (all but sites 1b, 4 and 8), the intensity of natural remanent magnetization (NRM) is high (Table I) compared to the ~ 10^{-6} emu cm^{-3} intensity common in continental red beds (Nagata, 1961) and the samples proved paleomagnetically useable (except at site 9 which was rejected largely because of

uncertainty in applying a tilt correction). At the three sites at which no oolitic hematite was detected (sites 1b, 4 and 8), the samples had low hematite content and weak NRM's compared to the rest of the sites and proved paleomagnetically unreliable.

We assume that a major part of the remanence at our paleomagnetically useable sites is carried by hematite ooliths (or detritus of hematite ooliths in the case of our finest-grained sandstones). This assumption was justified for one ironstone sample from site 1e by an experiment described later in which the magnetic moment of a thin slice of the ironstone was reduced by about a quarter on removing roughly a quarter of the hematite by physically removing the largest ooliths.

3. Test for isotropy of remanence acquisition

Howell et al. (1958, 1960) reported ~ 5% susceptibility anisotropy for the Red Mountain Formation ironstones at the one site that they studied on the southeast limb of the Birmingham anticlinorium. Normally, 5% susceptibility anisotropy is tolerable since it should not deflect remanence from the magnetizing field direction by more than ~ 3° (Stacey and Banerjee, 1974, p. 120). However, in the case of *red beds* with ~ 10% susceptibility anisotropy, Fuller (1963) sometimes found serious (~ 25°) deflection of remanence from the field direction. This was explained as due to the observed susceptibility anisotropy of the hematite crystals in such red beds being greatly reduced by interaction with a surrounding paramagnetic matrix. Such red beds could thus show small susceptibility anisotropy and yet be seriously anisotropic to remanence acquisition. This should be less of a problem in hematite ooliths since most of the hematite crystals should not be in a paramagnetic matrix.

To test for isotropy of remanence acquisition, a partial thermal remanent magnetization (pTRM) was given to four samples from site 1a and one each from sites 1c and 1e. In each case, the sample's bedding plane was horizontal and the Earth's field had an inclination of 71°. Vector subtraction of the remanence left after heating to 500°C and cooling in zero field from the net remanence after heating to 500°C and cooling in the Earth's field gave the pTRM acquired on cooling from 500°C. The pTRM direction seems to have been a little deflected toward the bedding plane (the plane of maximum susceptibility according to Howell et al., 1958, 1960) but still lies within 6° of the direction of the Earth's field in which it formed (Fig. 2). The ironstones will be assumed sufficiently isotropic for paleomagnetic use, though a small deflection toward the bedding plane is possible.

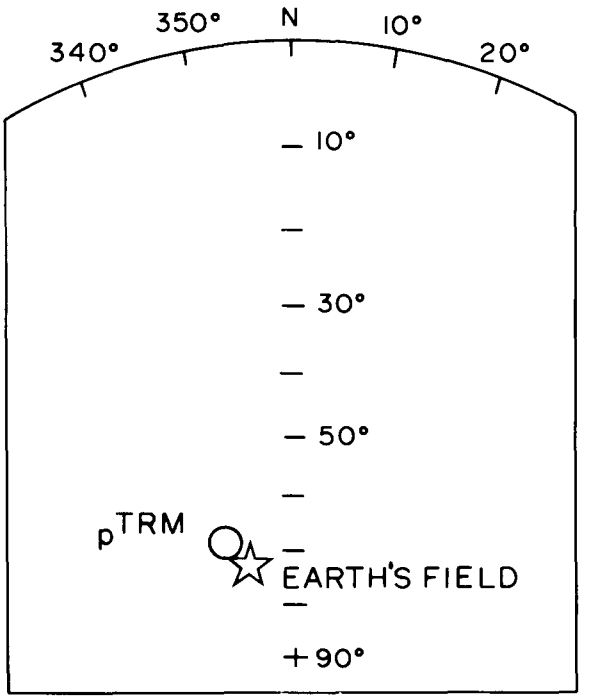


Fig. 2. Average direction and circle of 95% confidence for partial thermal remanent magnetization (pTRM) acquired on cooling from 500°C in the Earth's magnetic field (its direction shown with a star) for six Red Mountain Formation ironstone samples with their bedding planes horizontal. All directions point downward on an equal-angle stereographic projection.

4. Natural remanence and its demagnetization

All the samples were collected as oriented blocks from which cylindrical specimens (of ~ 2.3 cm diameter and ~ 1.9 cm height) were later cut. One specimen from each block sample was thermally

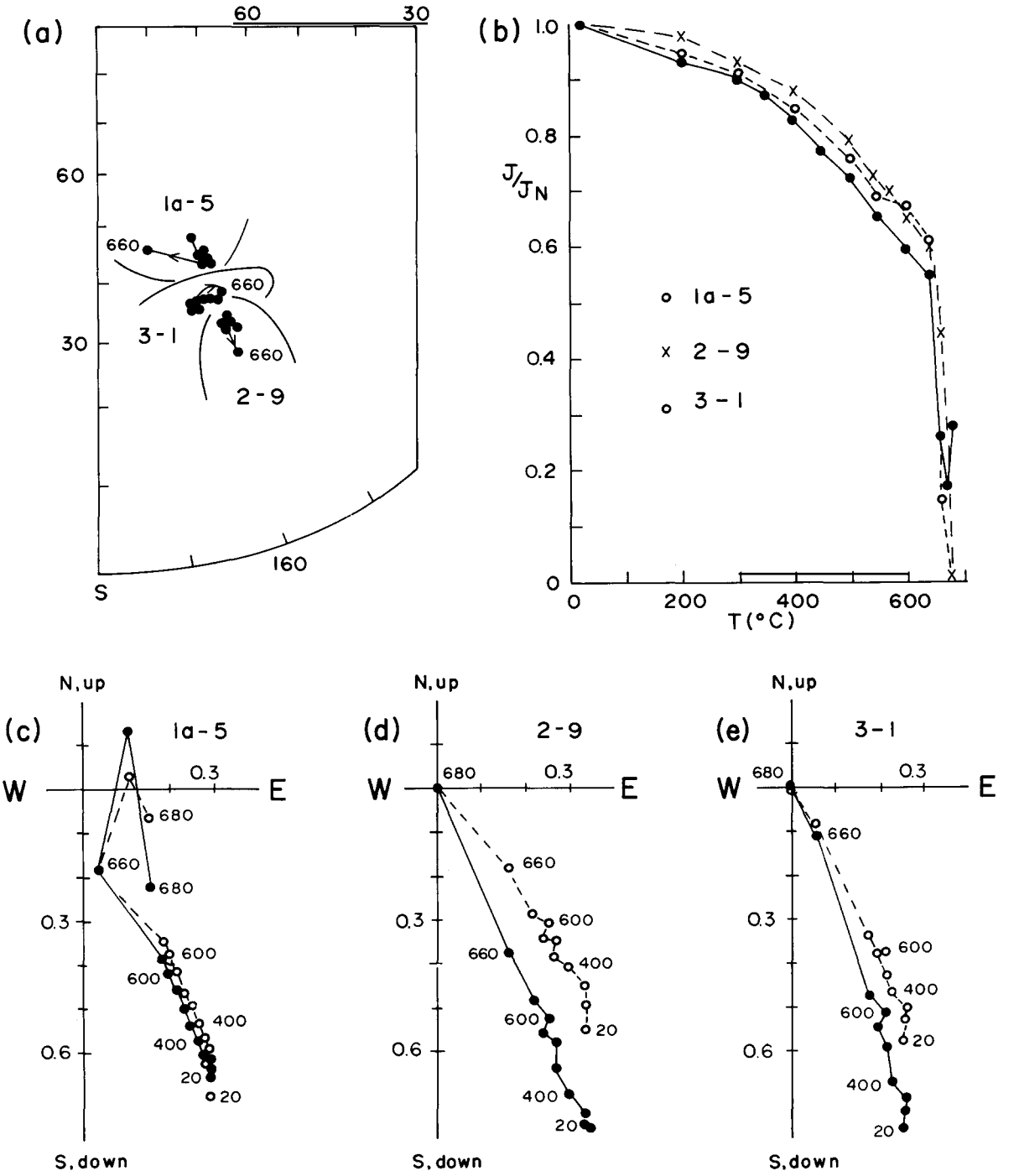


Fig. 3. Thermal demagnetization characteristics of typical Red Mountain Formation samples from stable sites on the southeast limb of the Birmingham anticlinorium. (a) Direction of remanence plotted on an equal-angle stereographic projection before tilt correction. Closed (open) symbols indicate downward (upward) inclinations. (b) Normalized intensity of remanence (J/J_N) as a function of temperature (T). (c), (d) and (e) Zijderveldt diagrams showing projection of magnetization vectors onto the horizontal plane (closed symbols) and onto the north–south vertical plane (open symbols). Axes are normalized to the total intensity of NRM. Samples 1a-1, 2-9 and 3-1 are from sites 1a, 2 and 3, respectively.

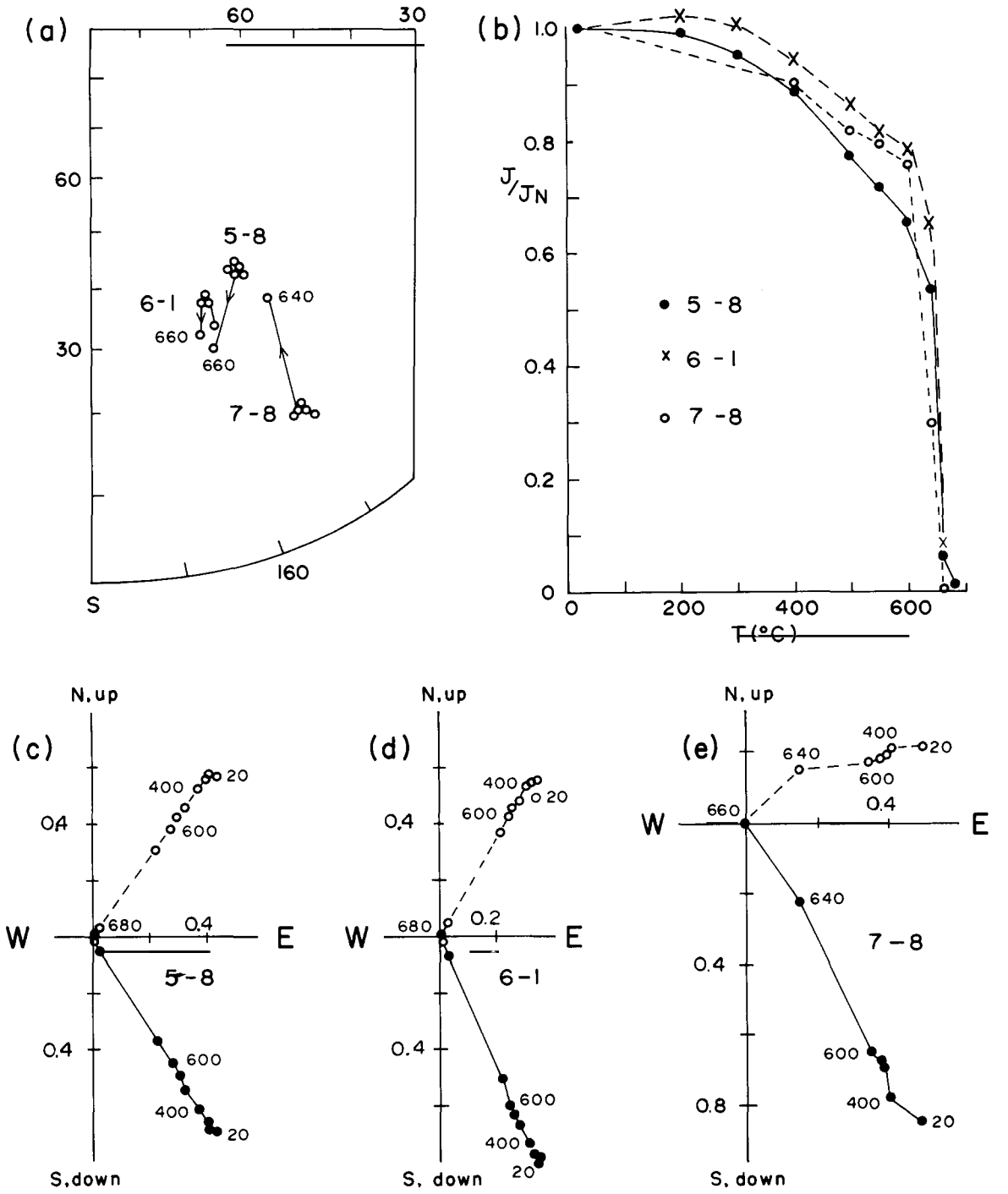


Fig. 4. Thermal demagnetization characteristics of typical Red Mountain Formation samples from stable sites on the northwest limb of the Birmingham anticlinorium. Directions are plotted before tilt correction. Symbols are the same as in Fig. 3.

demagnetized in steps, usually to peak temperatures of 200, 300, 400, 500, 550, 600, 640, 660 and 680°C. Usually, four to eight specimens were heated at a time in a Schonstedt TSD-1 demagnetizer with the peak temperature being maintained to within ± 5°C for 30 min and with the field at the specimens being nulled to $< 30\gamma$. Specimen magnetizations were measured initially and after each demagnetization step using a Schonstedt SSM-1 magnetometer.

At 7 of the 9 sampling sites on the SE limb of the Birmingham anticlinorium (sites 1a, 1c, 1d, 1e, 1f, 2 and 3) a consistent stable direction of magnetization is obtained on thermal demagnetization. Before correcting for the tilt of the beds, natural remanent magnetizations (NRM's) are directed to the SSE with inclinations of ~ 30–50°.

In the first steps (to 300°C) of the stepwise demagnetization procedure there are only minor directional changes, probably representing the elimination of small viscous components (Fig. 3a). Directions then remain quite stable, usually to 640°C or higher (600°C at sites 1c and 1d), as the intensities decrease slowly to ~ 60% of their NRM values (Fig. 3b). Above 640°C (600°C at sites 1c and 1d), the intensities fall rapidly and the directions begin to scatter, becoming random when only a few percent of the NRM intensity remains at 660 or 680°C.

On a Zijderveld plot of the horizontal and vertical components of remanence during demagnetization (Fig. 3c, d, e), data points between 300°C and 640 or 660°C (600°C at sites 1c and 1d) consistently fall on a straight line through the origin, indicating that only a single magnetic component is being erased over this temperature interval. Data points between 20 and 300°C may depart from the straight line due to the presence of a soft viscous component.

In samples from sites 1a, 1c and 1e, a rise in intensity is often observed in the final heating steps of 660 or 680°C (e.g., sample 1a-5 in Fig. 3b). This is thought to result from chemical alterations which produce magnetite in laboratory heating above 660°C. Such chemical alterations are not uncommon in studies of red beds (e.g., Buchan and Hodych, 1982a). However,, in the present case, only a small percentage of the samples in the collection are affected, and then only after their direction of magnetization has been defined. On a Zijderveld plot, such chemical effects produce a scatter of points which deviate drastically from the straight line through the origin (Fig. 3c).

On the NW limb of the Birmingham anticlinorium, stable directions of magnetization are isolated at 3 of the 5 sampling sites (sites 5, 6 and 7). Figure 4 shows typical examples. Before correction for tilt of the beds, NRM's are directed to the SSE with inclinations which usually vary between $+10$ and -35°.

Upon stepwise thermal demagnetization, directions typically move a few degrees to higher negative inclinations at 200 or 300°C (Fig. 4a), remain stable to 640 or 660°C (600°C at site 7) as the intensities fall slowly to between 40 and 90%. of their NRM values (Fig. 4b), then randomize as the intensities drop to a few percent at 660 or 680°C. On the NW limb, there is no indication of the chemical alterations that were observed in the high temperature thermal demagnetization steps at three sites from the SE limb.

Zijderveld diagrams for typical samples are shown in Fig. 4c, d, e. They illustrate the single component nature of the remanence at these sites.

Two sites (1b and 4) on the SE limb and two sites (8 and 9) on the NW limb do not yield consistent stable directions of magnetization (Fig. 5). Although directions at these sites are confined to the SE and SW quadrants, the behavior of individual samples is often variable.

At site 4 on the SE limb, magnetization directions continue to move on thermal demagnetization to temperatures of 540°C or higher (Fig. 5a). Even those which do eventually stabilize do not have a common endpoint. Similar results are observed at site 1b on the SE limb and at site 8 on the NW limb (Fig. 5b). At site 9, also on the NW limb (Fig. 5c), directions of magnetization follow a great circle path to a consistent endpoint which is not attained until 600°C in some samples. Here only 3 samples were collected from a small block which is tilted almost vertically. Hence it is difficult to rule out rotations in the plane of bedding. At all four sites, intensities show a wide variety of decay characteristics (Fig. 5d).

In some samples of sites 1b, 4, 8 and 9, where

directions swing along great circle paths, two components are clearly visible on a Zijderveld diagram. Figure 5e shows a particularly striking example from site 9. Unfortunately, however, the component erased between 20 and 600°C is not consistent among the samples measured from site 9.

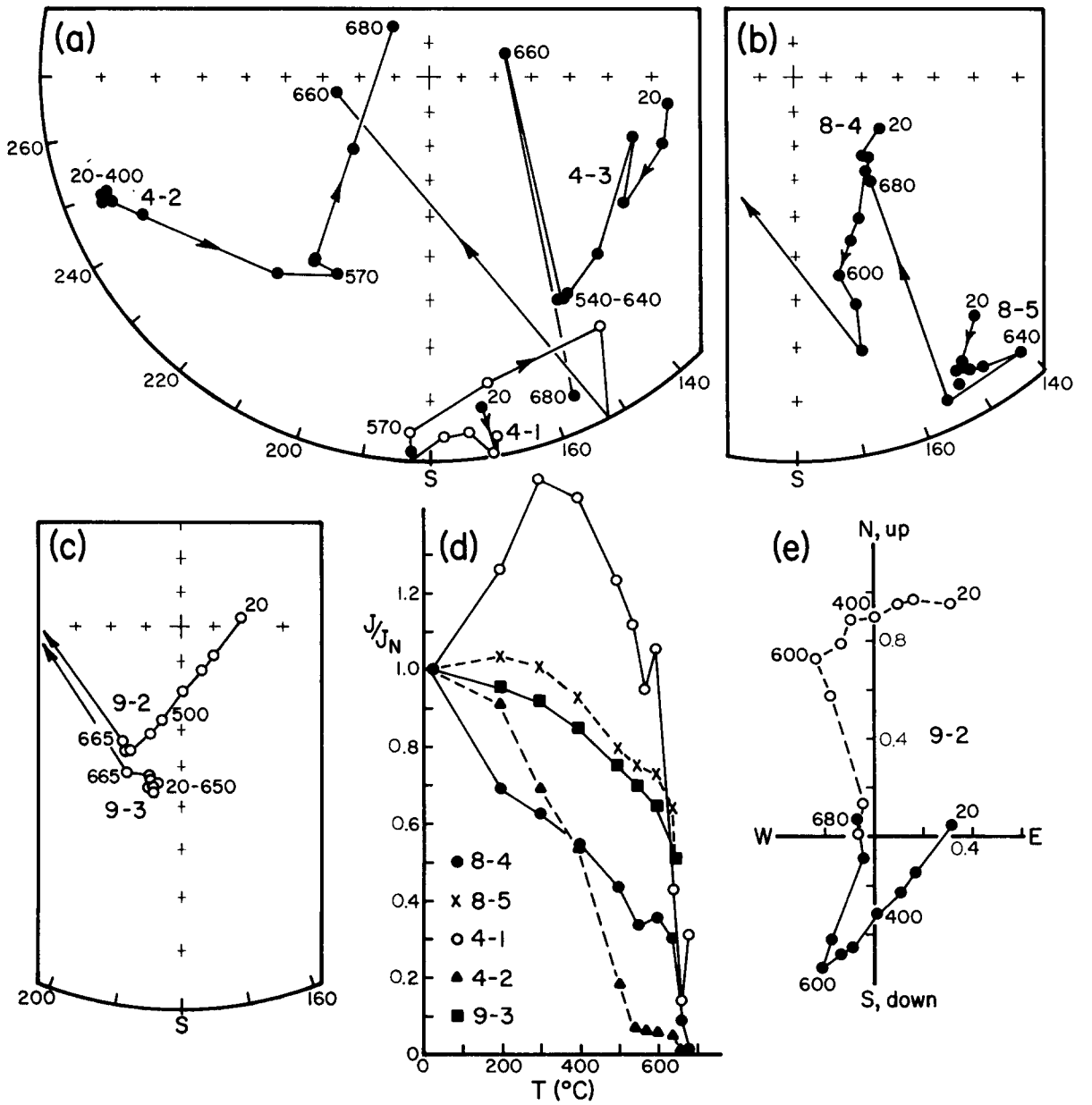


Fig. 5. Thermal demagnetization characteristics of typical Red Mountain Formation samples from "unstable" sites of the Birmingham anticlinorium. (a) Site 4 of the southeast limb. (b) Site 8 of the northwest limb. (c) site 9 of the northwest limb. (d) Normalized intensity results. (e) Zijderveld diagram of a two-component sample from site 9. All directions are shown before tilt correction. Symbols are the same as in Fig. 3.

TABLE II

Remanent magnetization directions of oolitic hematite-bearing sediments of Red Mountain Formation corrected for tilt of bedding

		Site mean remanence directions									
		Before demagnetization				After demagnetization to T°C					
Site	N	D	I	α_{95}	K	T°C	D	I	α_{95}	K	J/J_N
1a	5	150.2	23.5	2.7	827	640	150.2	20.3	2.5	929	0.66
1c	3	148.1	19.0	3.5	2176	600	149.0	16.2	3.4	1284	0.62
1d	11	151.2	21.4	3.1	217	600	150.6	17.1	3.7	156	0.60
1e	3	144.9	21.7	8.6	208	640	145.7	16.1	6.2	396	0.54
1f	3	147.0	22.1	11.2	122	640	151.7	18.5	9.5	171	0.45
2	6	155.9	16.2	8.9	58	640	155.6	13.1	9.0	57	0.59
3	4	150.1	23.7	14.7	40	640	149.2	22.1	10.1	84	0.63
5	8	140.8	21.0	4.6	148	640	138.3	20.2	4.5	153	0.66
6	9	154.6	34.0	9.4	31	640	151.3	25.4	9.4	31	0.48
7	8	157.1	22.4	6.0	85	600	154.4	20.1	6.3	77	0.72

N is the number of samples; D is the mean declination in degrees; I is the mean inclination in degrees; α_{95} is the half-angle of the cone of 95% confidence about the mean in degrees; K is the best estimate of precision; J/J_N is the fraction of remanence remaining after demagnetization to T°C.

Owing to the complex nature of the results at sites 1b, 4, 8 and 9, they are not used in the statistical calculations below.

Table II lists mean remanence directions before and after thermal cleaning for each of the ten paleomagnetically useable sites. Within-site scatter is invariably small to temperatures of 600–660°C (see α_{95}, the radius of the circle of 95% confidence, in Table II) and then increases dramatically as the intensities fall to a few percent of their NRM values in subsequent steps. The thermally cleaned mean directions of Table II are results obtained after heating to 640°C except at sites 1c, 1d and 7 where the 600°C results were used due to large increases in α_{95} at 640°C. These temperatures are sufficiently high to avoid contamination by viscous components.

5. The timing of remanence acquisition

For one ironstone sample from site 1e containing hematite mainly in abundant round ooliths, the remanent magnetic moment was measured for a thin polished disc about one oolith in thickness and 2.3 cm in diameter. Removal of the largest ooliths with a dental pick while viewing with a stereomicroscope did not significantly change the direction of magnetic moment but reduced its intensity by about a quarter corresponding roughly to the fraction of hematite estimated visually to have been removed. Besides suggesting that the hematite ooliths carry a major part of the remanence, this experiment suggests that the remanence of the ooliths was acquired after the ooliths ceased rolling on the sea floor accumulating layers of hematite or hematite precursor.

Cobbles of calcareous ironstone bearing oolitic hematite were collected from an intraformational conglomerate at the base of the Middle Red Mountain Formation (Chowns and McKinney, 1980) at site 1d. A great variation was observed in strike and dip of the internal bedding among the eleven cobbles collected (Fig. 6b). Nevertheless, the in situ remanence directions of the cobbles are all similar (Fig. 6a) and close to the in situ remanence directions of ironstone beds above and below (compare site 1d with sites 1a, 1c and 1e in Table II). This negative conglomerate test establishes that there was a time gap between deposition of the calcareous ironstone and its acquisition of remanence. But the negative conglomerate test does not establish the length of time gap except that it was at least long enough to permit the calcareous ironstone to be deposited, eroded into cobbles and incorporated into the conglomerate,

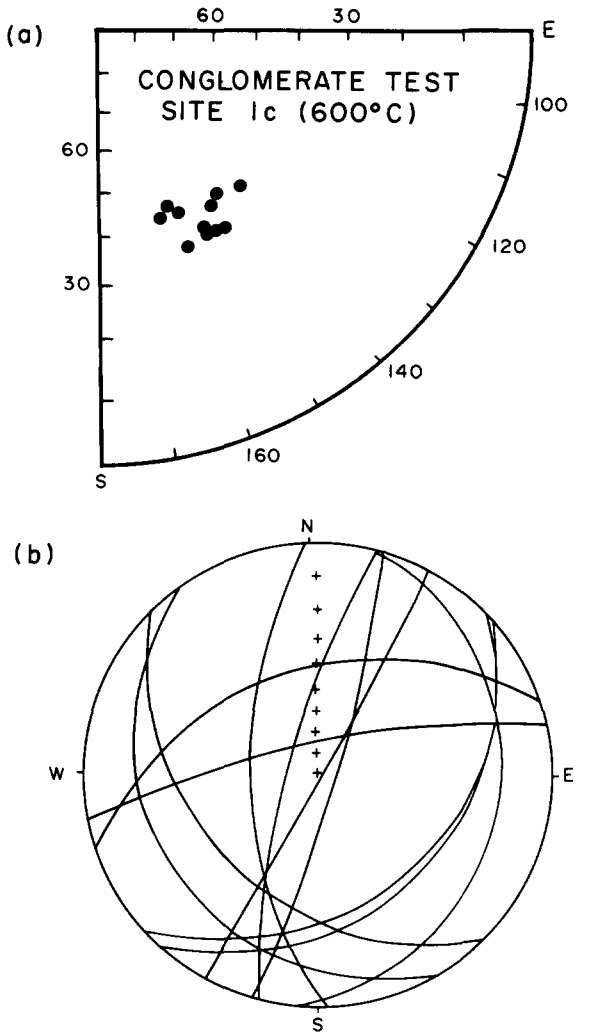


Fig. 6. (a) In situ remanence directions of eleven oolitic hematite bearing cobbles from a conglomerate in the Red Mountain Formation at site 1d after themal demagnetization to 600°C. (b) In situ orientation of individual bedding planes of the above eleven cobbles shown by arcs in stereographic projection on lower hemisphere.

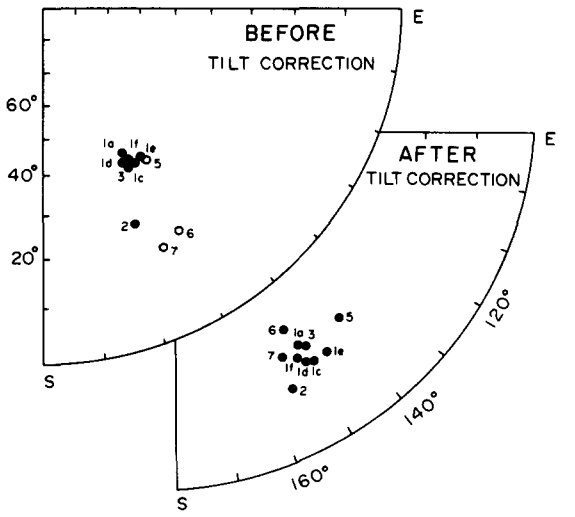


Fig. 7. Mean remanence directions of sites in the Red Mountain Formation after thermal demagnetization as in Table II. Before correcting for tilt of bedding, the remanence points downwards (closed circles) on the southeast limb of the Birmingham anticlinorium but upwards (open circles) on the northwest limb. After tilt correction, remanence directions on the two limbs do not differ significantly showing that remanence was acquired before folding.

all of which occurred during the Llandoverian. That is, the negative conglomerate test alone does not rule out the possibility that the remanence was acquired during the Silurian.

Before correcting for tilt of bedding, the mean remanence directions of sites on the southeast limb of the Birmingham anticlinorium are different from sites on the northwest limb (Fig. 7). After correcting for tilt of bedding, the remanence directions on the two limbs are not significantly different. The statistical test of McFadden and Jones (1981) showed that at the 95% level of confidence the mean direction of sites on the southeast limb (1a, 1c, 1d, 1e, 1f, 2 and 3) are significantly different from the mean direction of sites on the northwest limb (5, 6 and 7) before but not after tilt correction. This positive fold test demonstrates that remanence acquisition preceded the folding. Folding occurred after the Early Pennsylvanian judging by the magnitude of folding of Lower Pennsylvanian rocks along the Birmingham anticlinorium (Thomas and Neathery, 1980). Folding had probably occurred by the Early Permian because farther north in the Valley and Ridge Province, red beds often carry a post-folding remanence whose direction is closest to that of the Pennsylvanian or perhaps Early Permian field (Van der Voo, 1979).

After tilt correction and thermal cleaning, the ten site mean remanence directions of Table II yield a mean remanence direction for the Red Mountain Formation with $D = 149.6°$, $I = 19.0°$ ($\alpha_{95} = 3.44°$, $k = 198.2$). The corresponding

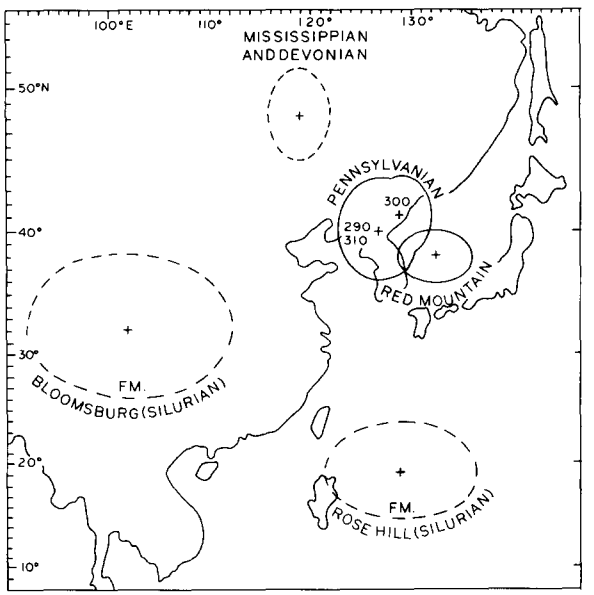


Fig. 8. The Red Mountain Formation paleopole shown at 38.0°N and 132.4°E with its oval of 95% confidence having semi-major and semi-minor axes $dm = 3.6°$ and $dp = 1.9°$ (giving unit weight to each of the thermally demagnetized site mean remanence directions of Table II). This paleopole is in good agreement with the mean 290, 300 and 310 Ma Pennsylvanian paleopoles for North America. It differs from the two most reliable Silurian paleopole determinations for cratonic North America and from the mean of Devonian and Mississippian paleopoles for cratonic North America.

paleopole falls at 38.0°N, 132.4°E ($dm = 3.6°$, $dp = 1.9°$). This paleopole is in good agreement (Fig. 8) with the well-defined mean Pennsylvanian paleopoles for North America (Irving and Irving, 1982). It differs from the well-defined mean Permian paleopoles for North America (Irving and Irving, 1982) and from the less well-defined mean Mississippian and Devonian paleopoles for cratonic North America (Irving, 1979). The Red Mountain Formation paleopole also differs from the few reliable Silurian paleopoles for cratonic North America—those for the Bloomsburg Formation (Roy et al., 1967) and for the Rose Hill Formation (French and Van der Voo, 1979). Hence, the direction of the Red Mountain Formation remanence suggests that it was acquired during the Pennsylvanian.

A Pennsylvanian acquisition of remanence is supported by the reverse polarity at all of the Red Mountain Formation sites. Reverse polarity is characteristic of almost all of the Pennsylvanian whereas normal or mixed polarity seems characteristic of the Mississippian, Devonian and Silurian (Irving and Pulliah, 1976; Roy and Morris, 1983). A Pennsylvanian pre-folding date for remanence acquisition in the Red Mountain Formation is also compatible with the negative conglomerate test and the positive fold test.

6. Mechanism of remanence acquisition

Oolitic hematite occurs at all ten of the paleomagnetically useable sites of the Red Mountain Formation. The remanence at all the useable sites is carried by hematite as shown by the blocking temperatures which are mostly in the 600–670°C range. The remanence is essentially single-component at all the useable sites and after tilt correction does not significantly differ from the direction expected of the Pennsylvanian. This suggests that the remanence was mostly acquired by a single mechanism during the Pennsylvanian and before folding.

A detrital remanent magnetization is ruled out for the calcareous ironstone cobbles at site 1d because the negative conglomerate test establishes a considerable time gap between calcareous ironstone deposition and remanence acquisition. Indeed a detrital remanent magnetization is unlikely at any of the useable sites since there was likely ~ 130 Ma between deposition and remanence acquisition judging by the direction and polarity of remanence.

A thermal remanent magnetization at any of the useable sites is also unlikely. The hematite has blocking temperature mostly above 600°C requiring temperatures of ~ 480°C or more to unblock according to the theoretical estimates of Pulliah et al. (1975). But the temperature of the Red Mountain Formation was probably never raised much above 250°C judging by the rank of nearby Pennsylvanian coals (Culbertson, 1964) interpreted after Hood et al. (1975). The color of conodonts in nearby Ordovician limestones (Harris et al., 1978) suggests even lower maximum temperatures when interpreted using the dry air

experimental Arrhenius plots of Epstein et al. (1977). However, heating in nature would likely have been in water which should raise the estimate of maximum temperature closer to that from coal rank (Epstein et al., 1977).

Having shown that detrital and thermal remanance are unlikely, we conclude that the Red Mountain Formation very likely carries a chemical remanent magnetization (CRM) residing in hematite. The CRM was likely the result of sediment diagenesis which Fairbridge (1983) defined as those physical and chemical changes induced in sediment during and after deposition at relatively low temperature ($< 200°C$) and pressure. Fairbridge (1983) divided diagenesis into three phases—a depositional and shallow-burial phase, a deep-burial phase and a post-uplift phase. The CRM of the Red Mountain Formation was likely acquired in the deep-burial phase of diagenesis. Certainly CRM acquisition during the depositional or shallow-burial phase of diagenesis can be discounted since the Red Mountain Formation was buried under at least 1 km of sediment (King, 1950) during Pennsylvanian remanence acquision. If CRM acquisition had occurred during the post-uplift phase of diagenesis, we would expect a post-folding remanence. We would also expect a post-folding remanence if CRM had been acquired during diagenesis induced by the large-scale post-deformational (telemagmatic?) heating postulated (Damberger, 1974) to explain large-scale post-deformational coal-rank increases. (This adds a fourth possible phase of diagenesis to the three of Fairbridge.)

Creer (1968) hypothesized that lateritization commonly produced secondary CRM in red beds during the Carboniferous and Permian when the northern continents lay at low latitudes. However, such lateritization can be classed as a diagenetic change of the post-uplift phase and would be expected to produce a post-folding remanence in the Red Mountain Formation. Pre-folding Pennsylvanian lateritization would have had to penetrate ~ 1 km thickness of Mississippian sediment to reach the Red Mountain Formation but these Mississippian sediments show no reddening.

TABLE III

Possible effects of diagenesis upon magnetic minerals and remanent magnetization of oolitic hematite ironstones

Phase of diagenesis	Expected physical and chemical conditions (partly after Fairbridge, 1983)	Expected effect on magnetic minerals and resulting remanence	
		Hypothesis I (throughout Phanerozoic)	Hypothesis II (for Early Paleozoic only)
Depositional and shallow-burial phase	Temperature $< 50°C$ pH ~ 8 in ocean but variable 7–9 below sediment surface	Goethite forms Resulting CRM unstable since blocking temperature $< 120°C$	Hematite forms from ferrihydrite. CRM in $> 0.03\ \mu m$ grains of hematite
Deep-burial phase *	Temperature 50°–200°C pH 8–9 increasing with depth	Goethite transforms to hematite at ~ 80°C Resulting CRM is prefolding but may greatly postdate deposition	Hematite may coarsen as temperature rises if pH $\gtrsim 8$ Prefolding CRM in hematite coarsened through $0.03\ \mu m$
Post-depositional telemagmatic(?) heating *	As above	Goethite transforms to hematite at ~ 80°C if no previous deep burial Resulting CRM is post folding	Hematite may coarsen as temperature rises if pH $\gtrsim 8$. Post folding CRM in hematite coarsened through $0.03\ \mu m$
Uplift phase	Temperature $< 50°C$ pH 7–9	Hematite may coarsen if pH < 8 No effect on CRM since hematite likely already in $> 0.03\ \mu m$ grains	Hematite may coarsen if pH < 8 Post folding CRM in hematite coarsened through $0.03\ \mu m$

* Either or both of these stages of diagenesis may be absent or the temperature may be high enough ($> 200°C$) to pass to low temperature metamorphism.

Perroud and Van der Voo (1984) seemed to suggest that the hematite in the Red Mountain iron ores was of secondary origin replacing calcite cement and fossil fragments (Cooper and Hagni, 1981) in the Late Carboniferous during diagenesis perhaps due to fluid migrations resulting from tectonism during the early phases of the Alleghenian orogeny. However, it would be difficult to explain why such fluid migrations did not replace calcite with hematite in the overlying Mississippian or underlying Ordovician limestones or why they did not redden the grey sandstones of the Red Mountain Formation or of the overlying Mississippian. Besides, much of the remanence seems to be carried by hematite ooliths whose high aluminum content (see below) makes it unlikely that they originated by hematite replacement of calcareous ooliths (Maynard, 1983, p. 58).

We have argued that a major part of the remanence in the red sediments of the Red Mountain Formation likely resides in oolitic hematite. It is also likely that the hematite in the ooliths was already present at deposition (Maynard, 1983) though perhaps as a hematite precursor mineral such as ferrihydrite or goethite. The oolitic hematite acquired CRM during the Pennsylvanian probably as a result of deep-burial diagenesis as argued above. We now consider two possible mechanisms for CRM acquisition during deep-burial diagenesis of oolitic hematite in the Red Mountain Formation. Table III compares the two hypotheses. The hematite replacing calcite cement or fossil fragments is assumed to have had a similar history to that of the hematite in the ooliths. The grey fine-grained sandstones of the Red Mountain Formation often carry a remanence closely resembling that of the ironstones except for being much weaker (Perroud and Van der Voo, 1984). This similarity may be due to the hematite in the grey sediments being in the form of detritus of hematite ooliths (which should not redden the sediments as much as hematite cement would).

7. Hypothesis I: CRM acquisition by transformation of original goethite to hematite

We have hypothesized (Hodych et al., 1983, 1984) that the Red Mountain ores originated as oolitic goethite which transformed to hematite and acquired a CRM when heated to ~ 80°C.

That most oolitic hematite ores have diagenetically transformed from original goethite is a popular hypothesis (James, 1966; Maynard, 1983). It offers a uniformitarian explanation for why hematite is much more common than goethite in the oldest (Early Paleozoic) oolitic ores whereas goethite predominates over hematite in the younger oolitic ores (James, 1966).

The chemistry, at least in the case of less mobile elements such as Fe and Al (Amstutz and Bubenicek, 1967), seems similar for Paleozoic oolitic hematite ores and younger oolitic goethite ores. For example, wt.% Al_2O_3/Fe averages 0.12 for the eleven Paleozoic oolitic hematite ore analyses and 0.15 for the fourteen younger oolitic goethite ore analyses listed by James (1966). The Red Mountain Formation ores fit this generalization, having Al_2O_3/Fe averaging 0.10 (Burchard and Butts, 1910).

Certainly the microstructure of Mesozoic goethite ooliths (Rohrlich, 1974) and that of Paleozoic hematite ooliths (Bhattacharyya and Kikimoto, 1982) is similar. Both consist of thin plates of ferric oxide several microns in diameter that are arranged at a tangent to the surface of the oolith in concentric layers. (However, Schoen (1962) noted that though goethite transforming to hematite should reduce its volume by about a quarter, shrinkage cracks in ooliths are rare in the Silurian oolitic hematite of New York State.)

Theoretically, Langmuir (1971, 1972) showed that while fine-grained ($< 1/10$ μm) goethite should be unstable relative to hematite under geologic conditions, a temperature of ~ 80°C should be required to transform coarse-grained goethite (> 1 μm) to hematite in 100% humidity. If the several micron diameter plates of goethite observed in ooliths (Rohrlich, 1974) are single crystals, the goethite should be coarse enough to remain stable below 80°C in the 100% humidity expected in the subsurface (to depths up to at least 3 km according to Chapman, 1981). However, the plates of goethite in ooliths may be made up of microcrystals ~ 0.05 μm across (Radczewski and Schneiderhöhn, 1962; Vernet, 1969). If so, the microcrystals are perhaps close-packed enough to

have the stability of coarse goethite. (The transformation from goethite to hematite probably occurs by dissolution and reprecipitation rather than by solid state dehydration according to Schwertmann and Taylor, 1977.) This would account for why oolitic goethites commonly remain stable for over 100 Ma.

If the Red Mountain Formation oolitic hematites did transform from goethites at ~ 80°C, one can readily explain why their remanence was acquired in the Pennsylvanian before folding. Assuming a normal geothermal gradient of 26°C km^{-1} (Brooks, 1981), burial under ~ 2.3 km of sediment would be required to attain 80°C. There is no evidence for this deep a burial before the Pennsylvanian since the combined thickness of the Devonian and Mississippian sediments is only ~ 1 km in the Birmingham district (King, 1950). However, at least another 1.5 km of Lower Pennsylvanian sediment was added before folding (King, 1950) giving deep enough burial to attain 80°C. Indeed, pre-folding heating was likely to be considerably higher than 80°C since Lower Pennsylvanian coal in the Birmingham area is typically bituminous near the high to medium volatile boundary (Culbertson, 1964). This rank of coal requires ~ 130°C if heated for 100 Ma or ~ 160° if heated for 10 Ma according to Hood et al. (1975). Much of this heating was likely due to deep burial (Culbertson, 1964) under a thick Pennsylvanian clastic wedge from the east (King, 1977) and deepest burial probably occurred before folding. Hence, it is plausible that 80°C was first reached in the Birmingham area during the Pennsylvanian before folding. This would have transformed goethite to single-domain (> 0.03 μm according to Banerjee, 1971) hematite producing a Pennsylvanian pre-folding CRM. That a stable CRM can result from the transformation of goethite to hematite was shown by Hedley (1968).

8. Hypothesis II: CRM acquisition by coarsening of original fine-grained hematite

Chukhrov (1974) suggested that scarcity of land plants in the pre-Devonian Paleozoic would have allowed iron to be transported to the sea as ferrihydrite rather than in organic complexes. Ferrihydrite, being unstable, would soon transform to hematite or to goethite but hematite would be favoured in the pH of ~ 8 typical of the ocean (Schwertmann and Taylor, 1977). This would explain the predominance of hematite over goethite in Early Paleozoic oolitic iron ores.

If the Red Mountain Formation ore did originate as hematite from a ferrihydrite precursor, the timing of remanence acquisition suggests that most of the original hematite was too fine-grained to acquire a stable remanence and remained so until the pre-folding Pennsylvanian. That is, the grain size remained near to or below the ~ 0.03 μm diameter (Banerjee, 1971) at which the superparamagnetic to single-domain transition occurs in hematite. Ferrihydrite occurs as very fine particles of ~ 0.01–0.005 μm diameter (Schwertmann and Taylor, 1977). Such particles could transform to hematite particles too fine to acquire CRM (although the ferrihydrite particles do have a strong tendency to aggregate and in laboratory experiments produce hematite crystals ~ 0.05 μm across (Fischer and Schwertmann, 1975) which may be too large to be superparamagnetic). It is possible that the hematite grains did not sufficiently coarsen to acquire stable CRM until heating during Pennsylvanian deep burial raised the very low dissolution rate of hematite. (The dissolution rate of hematite should roughly double with every 10°C rise in temperature judging by experiments with HCl by Sidhu et al., 1981. However, if pH rises from 8 to 9 during deep burial as suggested by Fairbridge, 1983, the low solubility of Fe^{3+} would be reduced from 6×10^{-18} to 6×10^{-21} mol l^{-1} according to Fairbridge and there might be no coarsening of hematite.) Once the grains had grown to stable single-domain grain size, their CRM direction should have been unaffected by further increases in grain size. (Grain size has mostly remained below the ~ 15 μm diameter at which, according to Banerjee, 1971, hematite becomes multidomain.)

A serious difficulty with this mechanism of CRM acquisition is that one would expect remanence to be acquired over a long period of time giving a complex remanence with some Silurian and Mississippian components as well as the main

Pennsylvanian component. However, small Silurian and Mississippian components would be difficult to detect because they would not greatly differ from the Pennsylvanian direction and could be of mixed polarity.

9. Testing whether Early Paleozoic oolitic hematites transformed from goethites at ~ 80°C

That hematite is much more common than goethite in Early Paleozoic oolitic iron ores whereas the reverse is true of younger ores can be explained either by following James (1966) as in hypothesis I or by following Chukhrov (1974) as in hypothesis II. Hypothesis I can be tested more easily since it predicts that all oolitic goethite ores have been heated to less than ~ 80°C whereas most oolitic hematite ores have been heated to more than ~ 80°C. This will be tested using depth of burial, vitrinite reflectance or rank of coal and conodont color alteration. A depth of burial of ~ 2.3 km for example is required to attain ~ 80°C assuming a normal geothermal gradient. Vitrinite reflectance or rank of coal will be used to estimate paleotemperature following Hood et al. (1975) and Bostick et al. (1978). Conodont color alteration index will also be used to estimate paleotemperature using the dry air Arrhenius plots of Epstein et al. (1977) but keeping in mind that these may underestimate paleotemperature as discussed earlier.

Hypothesis I predicts that some oolitic goethites may have survived from the Early Paleozoic by not having been heated to 80°C. One possible example is the Lower Ordovician oolitic goethite of Siljan, Sweden (Hessland, 1949) where the conodont color alteration index in Lower Ordovician carbonates is only 1–1.5 (Bergström, 1980) suggesting heating to less than 80°C. Another possible example is the Ordovician oolitic goethite of Mayville Wisconsin (Hawley and Beavan, 1934) where burial may never have been deep enough for heating to 80°C. Such examples support hypothesis I but are also explicable with hypothesis II if acidic waters are postulated causing ferrihydrite to preferentially transform to goethite rather than to hematite (Schwertmann and Taylor, 1977).

Hypothesis I predicts that most Early Paleozoic oolitic hematites have been heated to at least 80°C. This is plausible in view of their great age but we have not done a systematic test. Certainly the Ordovician oolitic hematite ores from Czechoslovakia listed by Zitzmann (1978) should have been heated far beyond 80°C since they are all described as occurring in schists or quartzites. The Silurian oolitic hematites of the eastern U.S.A. have likely been heated to 80°C. This has already been demonstrated for the Birmingham area and can be shown for the Silurian oolitic hematites of Pennsylvania, Ohio and West Virginia (Hunter, 1970) using the rank of overlying Pennsylvanian coal (Damberger, 1974). The Silurian oolitic hematites of New York State have likely been heated to 80°C since the conodont color alteration index for nearby Silurian through Middle Devonian carbonates varies from ~ 2.7 at Clinton to ~ 2.0 at Rochester comparable to the ~ 2.3 for Ordovician carbonates at Birmingham (Harris et al., 1978).

The most diagnostic test of hypothesis I is offered by its prediction that *no* oolitic goethites should have survived long heating at temperatures much above 80°C. It is plausible that *most* oolitic goethites have not been heated much above 80°C though we have done no systematic test. The famous Jurassic oolitic goethites of Lorraine may never have been buried deeply enough to be heated to 80°C. This is true of the Toarcian–Aalenian oolitic goethite (limonite) ores at the north end of the Lorraine basin in Luxembourg (Zitzmann, 1978) judging by vitrinite reflectance averaging ~ 0.6% in nearby Liassic shales (Hagemann, 1973). In northwest Germany, there are Jurassic, Cretaceous and Tertiary oolitic goethites (Simon, 1969, p. 17) most of which have not been raised to 80°C judging by the mostly < 0.6% vitrinite reflectance for the Jurassic Posidonia shale in this area (Teichmüller and Teichmüller, 1981, p. 522). However, there is one example from this area of a Jurassic (Malm) oolitic goethite–hematite ore (though with hematite probably predominant according to Thienhaus, 1969) which appears to have survived long heating to considerably more than 80°C. This is at Nammen (52°14'N, 9°1'E) where a vitrinite reflectance of 1.6% has been

reported nearby in the Malm (Teichmüller and Teichmüller, 1981, p. 517) suggesting a temperature of ~ 150°C if applied for 100 Ma or ~ 180°C if applied for 10 Ma (Hood et al., 1975).

Hence it may be that oolitic goethite is stable to considerably higher temperatures than 80°C. This is supported by a recent suggestion (Yapp, 1983) that the substitution of Al for Fe which is common in oolitic goethites (Schwertmann and Taylor, 1977) may considerably increase the thermodynamic stability of goethite with respect to hematite. A substantial increase in the temperature needed to transform goethite to hematite may make it difficult for hypothesis I to account for the absence of all but secondary goethite in the Silurian oolitic hematites of New York State (Schoen, 1962) some of which may not have been heated much past 80°C judging by conodont color.

The situation may sometimes be more complex than either hypothesis I or II suggests. For example, the Upper Devonian or Lower Carboniferous Wadi Shati oolitic ores of Libya are hematitic despite having a lignite seamlet between two of the ore beds (Turk et al., 1980); the lignite suggests that plants were common on the adjacent continent and that heating was to less than 80°C (Bostick et al., 1978). Fortunately, such examples seem uncommon and hence can be treated as exceptions to hypothesis I or II. The occurrence of goethite at Nammen may similarly be treated as an exception at present.

10. Testing whether remanences of other oolitic hematites support hypothesis I or II

We have seen that the timing of remanence acquisition in the Silurian oolitic hematites of Birmingham is most easily explained by hypothesis I even if substantially more than 80°C is required to

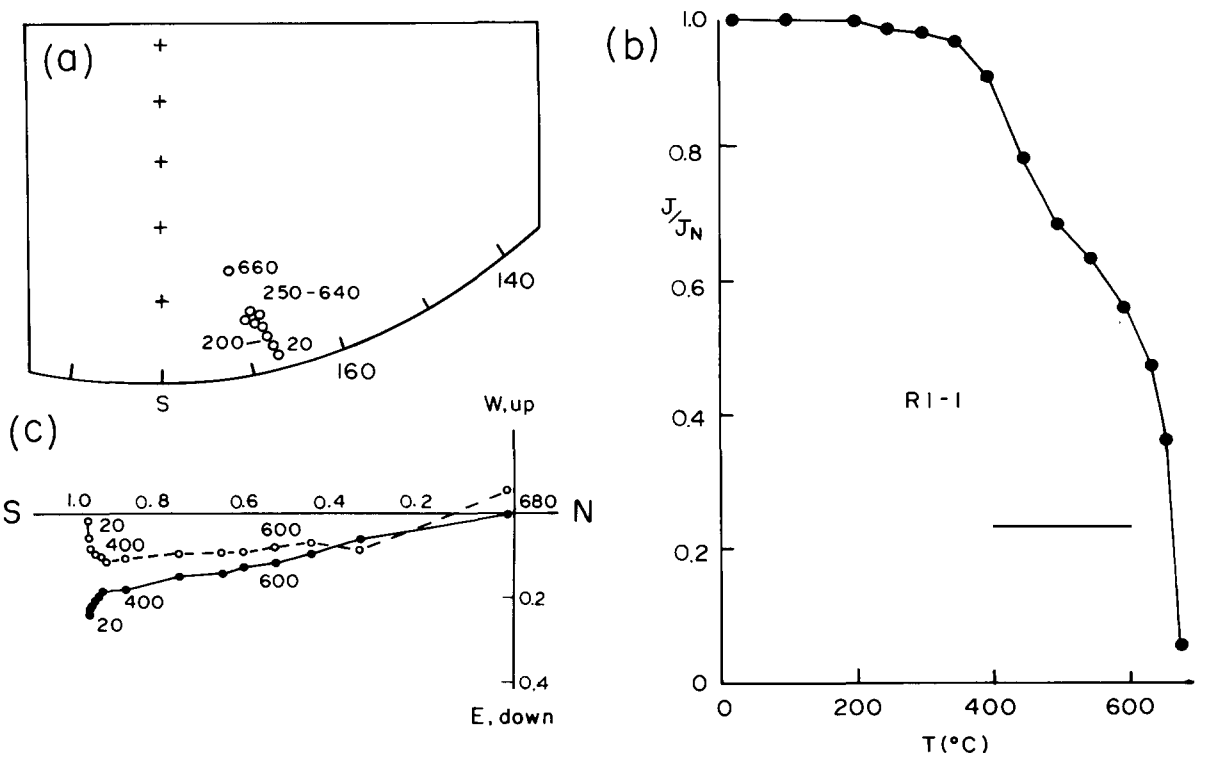


Fig. 9. Thermal demagnetization characteristics of a typical sample from oolitic hematite-bearing ironstones of the Silurian Clinton Group at Site R1 near Rochester, New York. Symbols are the same as in Fig. 3.

transform oolitic goethite to hematite though it is not inconsistent with hypothesis II. In the Ordovician oolitic hematite ores of Bell Island Newfoundland (Buchan and Hodych, 1982a), the remanence could be a CRM acquired by the mechanism of hypothesis I or II during Devonian orogenic heating but unfortunately the timing of remanence acquisition is not well established. The only other oolitic hematites reported on paleomagnetically are three Ordovician ores from Czechoslovakia (Andreeva et al., 1965) but these seem to have been metamorphosed and hence may carry a TRM.

This scarcity of oolitic hematites studied paleomagnetically leads us to report some preliminary results from the Silurian Clinton Group

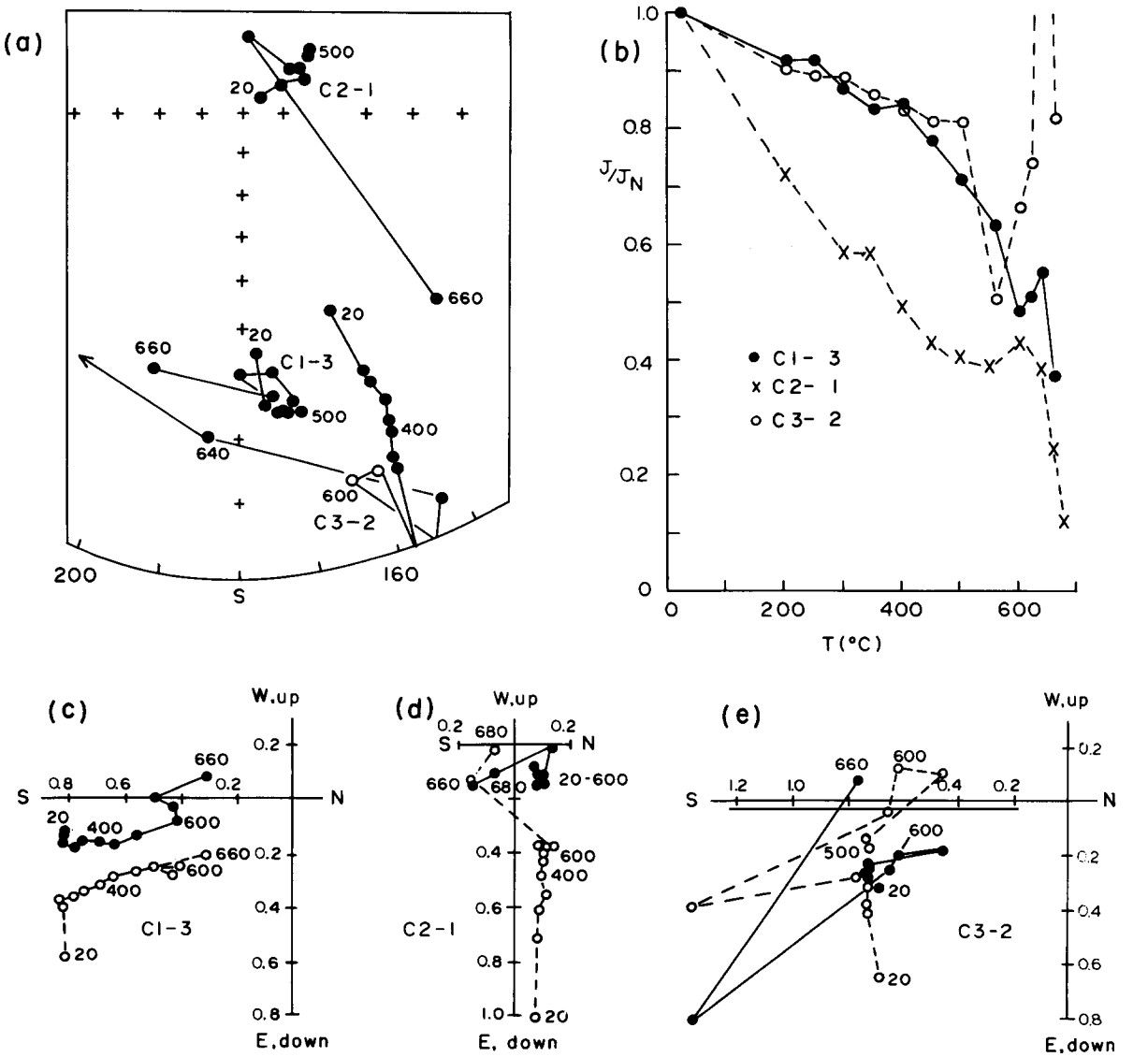


Fig. 10. Thermal demagnetization characteristics of one sample of oolitic hematite-bearing ironstone of the Silurian Clinton Group from each of sites C1, C2 and C3 near Clinton, New York. Symbols are the same as in Fig. 3.

of New York State. Block samples of oolitic hematite-bearing ironstone were collected at sites C1, C2 (both near 43°3.1'N, 75°21.3'W) and C3 (43°3.2'N, 75°21.9'W) near Clinton and at site R1 (43°11.1'N, 77°37.5'W) near Rochester. (See inset to Fig. 1.) Fourteen block samples were collected —three at Cl, three at C2, four at C3 and four at R1. One cylindrical specimen cut from each block sample was thermally demagnetized using the same steps as for most of the Red Mountain Formation specimens. The bedding was flat-lying except at site Cl (strike 180°, dip 8°W) where the remanence directions were tilt-corrected.

All four Rochester samples have high blocking temperatures (on average 60% of remanence remains after demagnetizing to 640°C) and strong NRM's (average intensity $= 4.4 \times 10^{-5}$ emu cm^{-3}). All four remanence directions are well-grouped and little affected by thermal demagnetization (Fig. 9 gives an example). Thermal demagnetization to 640° yields a mean $D = 171.7°$, $I = -10.2°$, $\alpha_{95} = 5.7°$, $k = 273$ for the four samples of site R1. The southerly declination and shallow inclination of remanence suggests that it may have been acquired during the Permian, though Siluro–Devonian or Carboniferous acquisition is also possible. Indeed, the remanence direction and polarity at site R1 agree with those of the Devonian Catskill red beds of New York State (Kent and Opdyke, 1978) which were likely magnetized before probable Permian folding (Van der Voo et al., 1979).

In contrast, the Clinton samples mostly have much lower blocking temperatures (on average 55% of remanence remains after demagnetizing to 500°C) and weaker NRM's (average intensity $= 1.7 \times 10^{-5}$ emu cm^{-3}). The remanence directions are not well-grouped at any of the Clinton sites before or after thermal demagnetization. Remanence often increases on demagnetization to above 500°C perhaps due to growth of magnetite from siderite decomposition. Though not well-grouped, site Cl specimens all possess a shallow southerly remanence direction which is reasonably stable to thermal demagnetization (one example is given in Fig. 10a and c) and may have been acquired during the Permian, Siluro–Devonian or Carboniferous. Site C3 specimens also tend to move towards a shallow southerly magnetization direction on demagnetization (one example is given in Fig. 10a). However, Zijderveld plots at site C3 (e.g., Fig. 10e) usually indicate only the elimination of a steep downward component before chemical alteration obscures the remaining remanence. Site C2 specimens (e.g., Fig. 10a and d) are magnetized steeply downwards suggesting a relatively recent remanence acquisition also in evidence at site C3.

Whether or not these paleomagnetic results support hypothesis I or II (modified to take account of local burial history) depends on when the remanence was acquired which is uncertain because of insufficient sampling and because there is probably no chance of a fold test. Although the burial history is not well known, most of the sediment load which once lay above the Clinton and Rochester ironstones was probably deposited during the Late Devonian by a clastic wedge derived from the east, analogous to the Pennsylvanian clastic wedge that provided most of the load above the Birmingham ironstones (Colton, 1970; King, 1977). Hence, hypothesis I or II modified for Clinton and Rochester would predict remanence acquisition either in the Late Devonian or in the Carboniferous since burial in the Devonian may not have been to greater than 1.5 km but another 0.5 km could have been added in the Carboniferous (extrapolating the isopachs of Colton, 1970). The shallow southerly remanence at Rochester and Clinton could have been acquired during the Late Devonian or the Carboniferous lending support to hypothesis I or II, (although hypothesis II runs into some difficulties since higher temperatures should have caused greater grain growth and hence higher blocking temperatures at Clinton rather than Rochester). However, the shallow southerly remanence could instead have been acquired during the Permian which would not have been predicted from hypothesis I or II since it is unlikely that a significant thickness of Permian sediment load was ever present (King, 1950). That is, Permian remanence acquisition would imply that timing of remanence acquisition was not primarily controlled by depth of burial or temperature. Some other factor would have to be involved—such as increased acidity of ground water speeding hematite dissolution and grain growth. Further work

may establish whether a Permian or a Devonian–Carboniferous remanence acquisition was more likely. Meanwhile, preliminary results are not inconsistent with hypothesis I or II.

11. Conclusions about remanence acquisition in oolitic hematites

We have shown that the oolitic hematite-bearing Red Mountain Formation likely carries a CRM acquired during deep-burial diagenesis. We have considered two hypotheses for how the CRM was acquired during deep-burial diagenesis. Hypothesis I follows from the view that most Phanerozoic oolitic hematite ironstones transformed from oolitic goethites whereas hypothesis II follows from the view that most Early Paleozoic oolitic hematites had an unstable ferrihydrite precursor (see summary in Table III). Both hypotheses encounter difficulties but remain plausible.

Hypothesis I—acquisition of CRM by transformation of original goethite to hematite at ~ 80°C —has the advantage of more easily explaining the timing of remanence acquisition in the Red Mountain Formation and possibly in the Clinton Formation. Hypothesis I is also easier to test and it does run into some difficulties. Oolitic goethites at Nammen in northwest Germany have not wholly transformed to hematite despite evidence of heating to ~ 150°C. This may be partly due to the Al-content of oolitic goethites raising the transformation temperature. If considerably more than 80°C is commonly required to transform oolitic goethite to hematite, it may be difficult to account for the absence of all but secondary goethite in the Silurian oolitic hematites of New York State (Schoen, 1962) since at least those at Rochester may not have been heated much beyond 80°C. Nevertheless, we prefer hypothesis I because it is easier to test and requires less special pleading than hypothesis II.

Hypothesis II involves CRM acquisition by coarsening of initial very fine-grained hematite during deep-burial heating. This mechanism of CRM acquisition can easily be modified by postulating a different initial hematite grain size or acidity of formation water to explain a wide variety of paleomagnetic results, making it difficult to test. In the case of the Red Mountain Formation, we have assumed that temperature was the main control on grain growth and that temperature was not high enough to cause sufficient hematite dissolution and grain growth for stable CRM until burial to more than 1 km depth in the Pennsylvanian raised the temperature beyond ~ 50°C. In the case of the Clinton Group, a higher temperature at Clinton has *not* caused greater grain growth than at Rochester judging by the lower blocking temperature at Clinton. However, this can be explained by postulating greater initial grain size or greater acidity of formation water at Rochester. If oolitic hematite ores have commonly originated from ferrihydrite, it will be difficult to predict when they were magnetized. If the original ferrihydrite grains were well-aggregated leading to large initial hematite grain size, CRM could be acquired during the deposition and early burial stage of diagenesis. If the initial hematite grain size was small, CRM could be acquired during the deep-burial stage of diagenesis (assuming pH did not rise from 8 to 9) when the rise in temperature increased the dissolution rate of hematite and speeded its coarsening to single-domain grain size. Diagenesis due to telemagmatic heating should have a similar effect but is likely to be post-folding. CRM could even be acquired during the uplift stage of diagenesis when there could be an influx of meteoric waters of lower pH (Fairbridge, 1983) which would increase the solubility of Fe^{3+} and speed the coarsening of hematite.

Interpretation of the paleomagnetism of Early Paleozoic oolitic hematites remains linked to the problem of whether they mostly transformed from oolitic goethites. It may be possible to solve this problem by further application of vitrinite reflectance and conodont color alteration to estimating the maximum temperatures experienced by oolitic goethites and hematites as briefly attempted above. It may also be possible to solve this problem paleomagnetically. For example, if the remanence of Early Paleozoic oolitic hematites was found to be commonly acquired in the deep-burial phase of diagenesis (pre-folding), it would support their having originated from goethite. On the other hand, if further work on identifying the fraction of

hematite carrying remanence showed that ooliths commonly did not carry remanence, it would suggest that their hematite formed and magnetized while the ooliths were still rolling on the sea floor.

12. Implications for remanence acquisition in continental red beds

This study supports the view (Turner, 1980) that diagenesis plays a dominant role in the magnetization of continental red beds. Larson and Walker (1975) already showed that diagenetic changes during relatively shallow (< 1 km) burial can cause CRM to be acquired over 1 or 2 Ma in continental red beds. There are many examples of continental red beds which carry a post-folding remanence (Turner, 1980); at least some of these are likely CRM's acquired during the uplift phase of diagenesis and others may be due to diagenesis during telemagmatic heating. Our study of oolitic hematites suggests that *deep-burial* diagenesis can cause acquisition of late but pre-folding CRM in continental red beds.

CRM acquisition in continental red beds could result from deep-burial heating speeding the intrastratal dissolution of Fe-silicate grains and producing hematite (analogous to hypothesis II for oolitic hematites). This mechanism could be most effective in continental red sandstones since much of their hematite pigment is thought to originate from intrastratal dissolution of Fe-silicate grains (Larson and Walker, 1975). Deep-burial heating might also transform any goethite present to hematite producing CRM (analogous to hypothesis I for oolitic hematites). This mechanism could be most effective in continental red mudstones since much of their hematite pigment might come from a goethite precursor (Hubert and Reed, 1978; Turner, 1980).

The remanence of red beds—continental or marine—might thus be acquired through deep-burial diagenesis long after deposition and yet *before folding*. Hence red bed paleomagnetism must be interpreted cautiously. Even a positive fold test may not prove that remanence was acquired close to the time of deposition. For example, the Mauch Chunk red beds and the Catskill red beds are the only Mississippian and Devonian rocks, respectively, from cratonic North America for which there are positive fold tests (Roy et al., 1983). Assuming these remanences to have been acquired close to the time of deposition provides important support for the Kent and Opdyke (1978, 1979) hypothesis that the northeastern Appalachians were located some 15 to 20° south of their present location relative to cratonic North America during the Mississippian and Devonian. However, since the Mauch Chunk and Catskill red beds were likely folded during the Permian, these red beds could carry a Pennsylvanian remanence in which case they would not support the Kent and Opdyke hypothesis (Roy et al., 1983). The present paper provides a mechanism whereby such a late pre-folding remanence may have been acquired (Irving and Strong, 1984); the Mauch Chunk and Catskill red beds may carry a CRM acquired during Pennsylvanian deep burial diagenesis. The Mauch Chunk lies at the top of the Mississippian and thus could well have escaped deep-burial diagenesis until thick Pennsylvanian sediments were added (> 1 km in the area of the fold test, Colton, 1970). The Catskill red beds were deeply buried in the Carboniferous in both areas studied paleomagnetically—New York State (Friedman and Sanders, 1982) and Pennsylvania (Colton, 1970)—whereas post-Carboniferous sediment is thought to have been thin (King, 1950).

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